

RaceDay API Endpoint Plan

1. API Overview

RaceDay is a RESTful API for managing road running, walking and cycling events in South Africa.

The API supports two main user roles:

- **Organiser** – creates and manages events and categories, views participant enrolments and records results.
- **Participant** – manages their profile, views events and categories, enrolls in events and views their own results.

Authentication is required for protected endpoints. The API should use an access token after successful login. The authenticated user's role is used to control access to organiser and participant functions.

2. Authentication Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new RaceDay user account and assigns a role.	None	{ firstName, lastName, email, password, phone, roleId }	201 Created – user created. 400 Bad Request – invalid/missing data. 409 Conflict – email already exists.
POST	/api/auth/login	Authenticates a user and returns an access token.	None	{ email, password }	200 OK – token and basic user information returned. 400 Bad Request – invalid request. 401 Unauthorized – incorrect email/password.

3. User Profile Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	<code>/api/profile</code>	Returns the profile of the currently logged-in user.	Organiser / Participant	None	200 OK – profile returned. 401 Unauthorized – user is not logged in. 404 Not Found – profile does not exist.
PUT	<code>/api/profile</code>	Updates the currently logged-in user's profile information.	Organiser / Participant	<code>{ firstName, lastName, email, phone }</code>	200 OK – profile updated. 400 Bad Request – invalid data. 401 Unauthorized – not logged in. 409 Conflict – email already belongs to another user.

4. Event Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	<code>/api/events</code>	Returns a list of available RaceDay events.	None	None	200 OK – list of events returned.
GET	<code>/api/events/{id}</code>	Returns details for one specific event.	None	None	200 OK – event returned. 404 Not Found – event does not exist.
POST	<code>/api/events</code>	Creates a new running, walking or cycling event. The logged-in organiser becomes the event organiser.	Organiser	<code>{ eventName, description, eventDate, location, distance, eventType }</code>	201 Created – event created. 400 Bad Request – invalid event data. 401 Unauthorized – not logged in. 403 Forbidden – user is not an organiser.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/events/{id}	Updates an event belonging to the logged-in organiser.	Organiser	{ eventName, description, eventDate, location, distance, eventType }	200 OK – event updated. 400 Bad Request – invalid data. 401 Unauthorized – not logged in. 403 Forbidden – organiser does not own event. 404 Not Found – event does not exist.
DELETE	/api/events/{id}	Deletes an event belonging to the logged-in organiser.	Organiser	None	204 No Content – event deleted. 401 Unauthorized – not logged in. 403 Forbidden – organiser does not own event. 404 Not Found – event does not exist. 409 Conflict – event cannot be deleted because it has existing enrolments/results.

Valid Event Types

The `eventType` value should support:

- Run
- Walk
- Cycle

The distance should be stored as a numeric value, for example `10.00` for a 10 km event.

5. Category Endpoints

Categories belong to a specific event. They can represent age-based or distance-based participation categories.

Examples include:

- Under 20
- Senior
- 10 km

- 21 km
- 42.2 km

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	<code>/api/events/{eventId}/categories</code>	Returns all categories belonging to a specific event.	None	None	200 OK – categories returned. 404 Not Found – event does not exist.
GET	<code>/api/categories/{id}</code>	Returns details of one category.	None	None	200 OK – category returned. 404 Not Found – category does not exist.
POST	<code>/api/events/{eventId}/categories</code>	Creates a new category for an event owned by the logged-in organiser.	Organiser	<code>{ categoryName, description, minAge, maxAge, minDistance, maxDistance, sortOrder }</code>	201 Created – category created. 400 Bad Request – invalid category data. 401 Unauthorized – not logged in. 403 Forbidden – organiser does not own event. 404 Not Found – event does not exist.
PUT	<code>/api/categories/{id}</code>	Updates a category belonging to an event managed by the logged-in organiser.	Organiser	<code>{ categoryName, description, minAge, maxAge, minDistance, maxDistance, sortOrder }</code>	200 OK – category updated. 400 Bad Request – invalid data. 401 Unauthorized – not logged in. 403 Forbidden – organiser does not own the event. 404 Not Found – category does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
DELETE	/api/categories/{id}	Deletes a category belonging to an event managed by the logged-in organiser.	Organiser	None	204 No Content – category deleted. 401 Unauthorized – not logged in. 403 Forbidden – organiser does not own event. 404 Not Found – category does not exist. 409 Conflict – category has existing enrolments.

6. Event Enrolment Endpoints

An enrolment connects a participant to an event and the category they selected.

This prevents the participant from simply registering for an event without selecting a category.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	<code>/api/events/{eventId}/enrolments</code>	Enrols the logged-in participant in an event using the selected category.	Participant	<code>{ categoryId }</code>	201 Created – enrolment created. 400 Bad Request – invalid category/event information. 401 Unauthorized – not logged in. 403 Forbidden – user is not a participant. 404 Not Found – event or category does not exist. 409 Conflict – participant is already enrolled in the event.
GET	<code>/api/profile/enrolments</code>	Returns all events the logged-in participant has enrolled in.	Participant	None	200 OK – enrolment list returned. 401 Unauthorized – not logged in.
GET	<code>/api/events/{eventId}/enrolments</code>	Returns all participant enrolments for an event managed by the logged-in organiser.	Organiser	None	200 OK – enrolments returned. 401 Unauthorized – not logged in. 403 Forbidden – organiser does not own event. 404 Not Found – event does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/enrolments/{id}	Returns details of one enrolment. Participants may only access their own enrolment; organisers may access enrolments for their own events.	Organiser / Participant	None	200 OK – enrolment returned. 401 Unauthorized – not logged in. 403 Forbidden – user does not have access. 404 Not Found – enrolment does not exist.
DELETE	/api/enrolments/{id}	Allows a participant to cancel their own enrolment.	Participant	None	204 No Content – enrolment cancelled. 401 Unauthorized – not logged in. 403 Forbidden – enrolment belongs to another participant. 404 Not Found – enrolment does not exist. 409 Conflict – enrolment can no longer be cancelled.

7. Result Endpoints

Results are recorded after an event. A result belongs to an enrolment, meaning the participant must first be enrolled in the event.

The result records the participant's finish time and finishing position.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Records a participant's finish time and finishing position for an enrolment.	Organiser	{ finishTime, finishPosition }	201 Created – result recorded. 400 Bad Request – invalid result data. 401 Unauthorized – not logged in. 403 Forbidden – organiser does not own the event. 404 Not Found – enrolment does not exist. 409 Conflict – a result already exists for the enrolment.
GET	/api/profile/results	Returns the logged-in participant's result history.	Participant	None	200 OK – participant results returned. 401 Unauthorized – not logged in.
GET	/api/enrolments/{enrolmentId}/result	Returns the result associated with a specific enrolment.	Organiser / Participant	None	200 OK – result returned. 401 Unauthorized – not logged in. 403 Forbidden – user does not have access. 404 Not Found – enrolment/result does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	<code>/api/events/{eventId}/results</code>	Returns results for participants in an event managed by the logged-in organiser.	Organiser	None	200 OK – event results returned. 401 Unauthorized – not logged in. 403 Forbidden – organiser does not own event. 404 Not Found – event does not exist.
PUT	<code>/api/results/{id}</code>	Corrects or updates an existing result recorded by the organiser.	Organiser	<code>{ finishTime, finishPosition }</code>	200 OK – result updated. 400 Bad Request – invalid result data. 401 Unauthorized – not logged in. 403 Forbidden – organiser does not own event. 404 Not Found – result does not exist.

8. HTTP Status Code Summary

The API will use standard HTTP response codes so that clients can understand whether a request succeeded or failed.

Status Code	Meaning	Example
200 OK	Request completed successfully.	Getting an event or updating a profile.
201 Created	A new resource was successfully created.	Registering a user or creating an event.
204 No Content	Request succeeded with no response body required.	Deleting an event or enrolment.

Status Code	Meaning	Example
400 Bad Request	Request contains invalid or missing information.	Invalid event date or missing required field.
401 Unauthorized	Authentication is required or credentials/token are invalid.	Accessing a protected endpoint without logging in.
403 Forbidden	User is logged in but does not have permission.	Participant attempting to create an event.
404 Not Found	Requested resource does not exist.	Requesting an event with an invalid ID.
409 Conflict	Request conflicts with existing data or the current state.	Participant trying to enrol twice in the same event.

9. Role and Security Rules

The API must enforce role-based access.

Public access

The following can be accessed without logging in:

- User registration
- User login
- View all events
- View a specific event
- View event categories

Organiser access

An authenticated Organiser can:

- View and update their own profile.
- Create events.
- Update their own events.
- Delete their own events.
- Create categories for their own events.
- Update/delete categories belonging to their own events.
- View enrolments for their own events.
- Record participant results.
- View results for their own events.
- Correct existing results.

An Organiser must not be able to modify another Organiser's events.

Participant access

An authenticated Participant can:

- View and update their own profile.
- View available events.
- View event categories.
- Enrol in events.
- View their own enrolments.
- Cancel their own enrolments where allowed.
- View their own result history.

A Participant must not be able to create, update or delete events/categories or record results.

10. Validation Rules

The API should validate important input before saving data.

Examples include:

- First name and last name cannot be empty.
 - Email addresses must be valid and unique.
 - Password must be supplied during registration.
 - Event name cannot be empty.
 - Event date must be a valid date.
 - Event distance must be greater than zero.
 - Event type must be `Run`, `Walk` or `Cycle`.
 - Category must belong to the selected event.
 - A participant cannot enrol twice in the same event.
 - A result must belong to an existing enrolment.
 - Finishing position must be a positive number.
 - Finish time must be a valid time value.
 - An organiser can only manage their own events.
 - A participant can only view or modify their own participant information.
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11. Endpoint Coverage

The planned API covers all RaceDay functional requirements:

Functional Requirement	Planned Endpoints
Authentication	<code>/api/auth/register</code> , <code>/api/auth/login</code>
User Profile	<code>/api/profile</code>
Events	<code>/api/events</code> , <code>/api/events/{id}</code>

Functional Requirement	Planned Endpoints
Categories	<code>/api/events/{eventId}/categories</code> , <code>/api/categories/{id}</code>
Event Enrolments	<code>/api/events/{eventId}/enrolments</code> , <code>/api/profile/enrolments</code> , <code>/api/enrolments/{id}</code>
Results	<code>/api/enrolments/{enrolmentId}/result</code> , <code>/api/profile/results</code> , <code>/api/events/{eventId}/results</code> , <code>/api/results/{id}</code>
Role-based access	Applied to all protected endpoints
Validation/failure handling	400, 401, 403, 404 and 409 responses planned

12. RESTful Design Principles

The RaceDay API uses RESTful resource-based routes rather than action-based routes where possible.

Examples:

- `GET /api/events` retrieves events.
- `POST /api/events` creates an event.
- `PUT /api/events/{id}` updates an event.
- `DELETE /api/events/{id}` deletes an event.
- `GET /api/events/{eventId}/categories` retrieves categories belonging to an event.

The API uses HTTP methods according to their intended purpose and uses resource IDs to identify specific records.

This endpoint plan will be used as the specification for the Part 2 C# RESTful API implementation. Any changes made during implementation should be documented and justified so that the API remains consistent with the approved plan.